MATHS MANTHAN
• TUITION CLASSES •
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Subject: Mathematics | **Periodic Test Paper 12 (Cumulative)** | **Time:** 60 Minutes
Class: Class X (CBSE) | **Chapter 5: Arithmetic Progressions** | **MM: 30**

Student Name: _____ **Roll No.:** _____ **Date:** _____

General Instructions:

- This question paper consists of **12 questions** divided into **Five Sections**: A, B, C, D, and E.
- **Section A** comprises **5 questions** of **1 mark** each (Q1–Q4 MCQs, Q5 Assertion-Reason).
- **Section B** comprises **2 questions** (Q6–Q7) of **2 marks** each.
- **Section C** comprises **2 questions** (Q8–Q9) of **3 marks** each.
- **Section D** comprises **2 questions** (Q10–Q11) of **5 marks** each.
- **Section E** comprises **1 Applied Case Study** (Q12) of **5 marks** with internal choice in sub-part (iii).
- All questions are compulsory. Use of calculators is strictly prohibited.

SECTION A: Multiple Choice & Assertion-Reason Questions (5 × 1 = 5 Marks)

Q1. The 30th term of the AP $10, 7, 4, \ldots$ is: [1]

(A) 97 (B) 77 (C) -77 (D) -87

Q2. The middle term of the finite AP $213, 205, 197, \ldots, 37$ is: [1]

(A) 125 (B) 117 (C) 121 (D) 129

Q3. Find the sum of all three-digit natural numbers which are divisible by 7: [1]

(A) 70336 (B) 70350 (C) 70314 (D) 70320

Q4. The sum of the first 20 odd natural numbers is: [1]

(A) 100 (B) 200 (C) 400 (D) 420

Q5. **Assertion (A):** In an AP with first term a and common difference d, the nth term from the end is given by $l - (n-1)d$.
Reason (R): Counting backwards from the last term l, the common difference becomes $-d$. [1]

(A) Both (A) and (R) are true, and (R) is correct explanation.
(B) Both (A) and (R) are true, but (R) is not correct explanation.
(C) Assertion (A) is true, but Reason (R) is false.
(D) Assertion (A) is false, but Reason (R) is true.

SECTION B: Very Short Answer Questions (2 × 2 = 4 Marks)

Q6. Find the 4th term of the AP whose sum of first n terms is given by $S_n = 2n^2 + 3n$. [2]

Q7. Find the sum of the first 20 terms of an AP in which the common difference $d = 7$ and the 20th term is 143. [2]

SECTION C: Short Answer Questions (2 × 3 = 6 Marks)

Q8. The 8th term of an AP is equal to three times its 2nd term, and its 12th term exceeds twice its 5th term by 1. Find the AP. [3]

Q9. Find the sum of all natural numbers between 250 and 1000 which are exactly divisible by 3. [3]

SECTION D: Long Answer / HOTS Questions (2 × 5 = 10 Marks)

Q10. If S_n denotes the sum of first n terms of an AP, prove that:

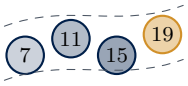
$$S_{12} = 3(S_8 - S_4)$$

[5]

Q11. The sum of the first 7 terms of an AP is 49 and that of the first 17 terms is 289. Find the sum of its first n terms, and hence calculate S_{25}. [5]

SECTION E: Applied Case Study (1 × 5 = 5 Marks)

Q12. CASE STUDY: Decorative Stepping Stones in Landscaped Walkway



A landscape architecture firm is contracted to lay out a paved promenade across a botanical garden.
The walkway incorporates circular numbered stepping stones engraved with sequential serial numbers that follow an arithmetic progression. The sequence of numbers on consecutive stones begins with $7, 11, 15, 19, \ldots$.
The pathway features 40 stones in total.

Based on the above architectural installation, answer the following questions:

(i) What number is engraved on the 25th stepping stone? [1]

(ii) Which stepping stone in the promenade is engraved with the number 115? [1]

(iii) Calculate the sum of all serial numbers engraved on the 40 stepping stones (S_{40}). [3]

OR

(iv) If a shorter segment consisting of only 31 stepping stones is considered, find the number engraved on the central (middlemost) stone. [3]

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★★★ END OF QUESTION PAPER · ALL THE BEST ★★★